



AE4238: Aero Engine Technology

Flight Performance and Propulsion

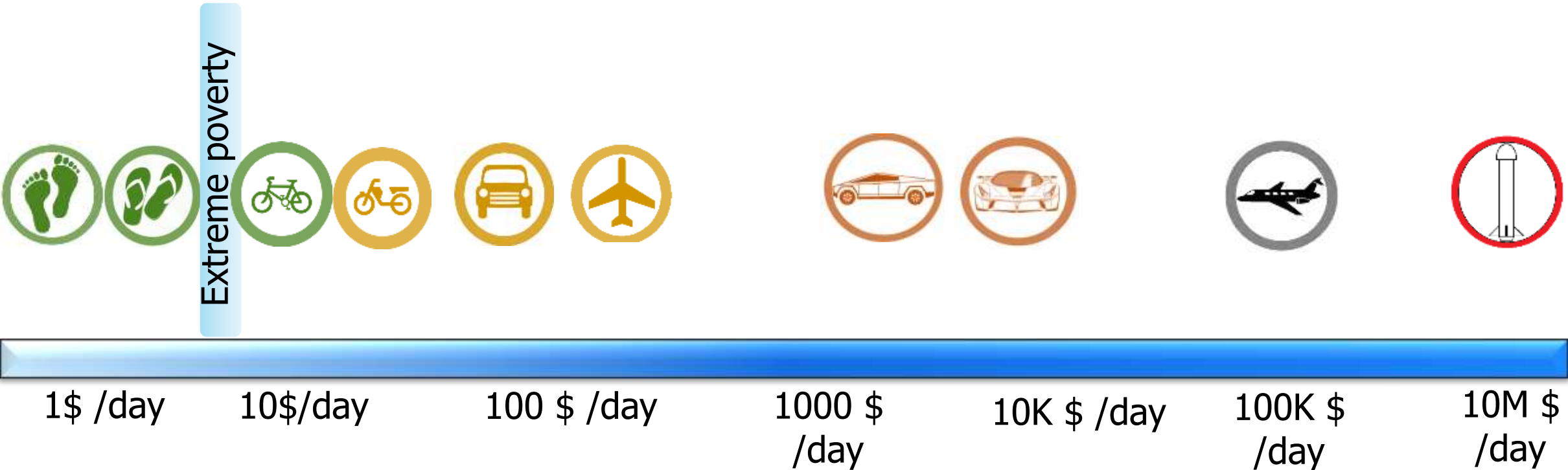
Prof. Dr. Arvind Gangoli Rao & Dr. Feijia Yin

The world wide web of Civil Aviation



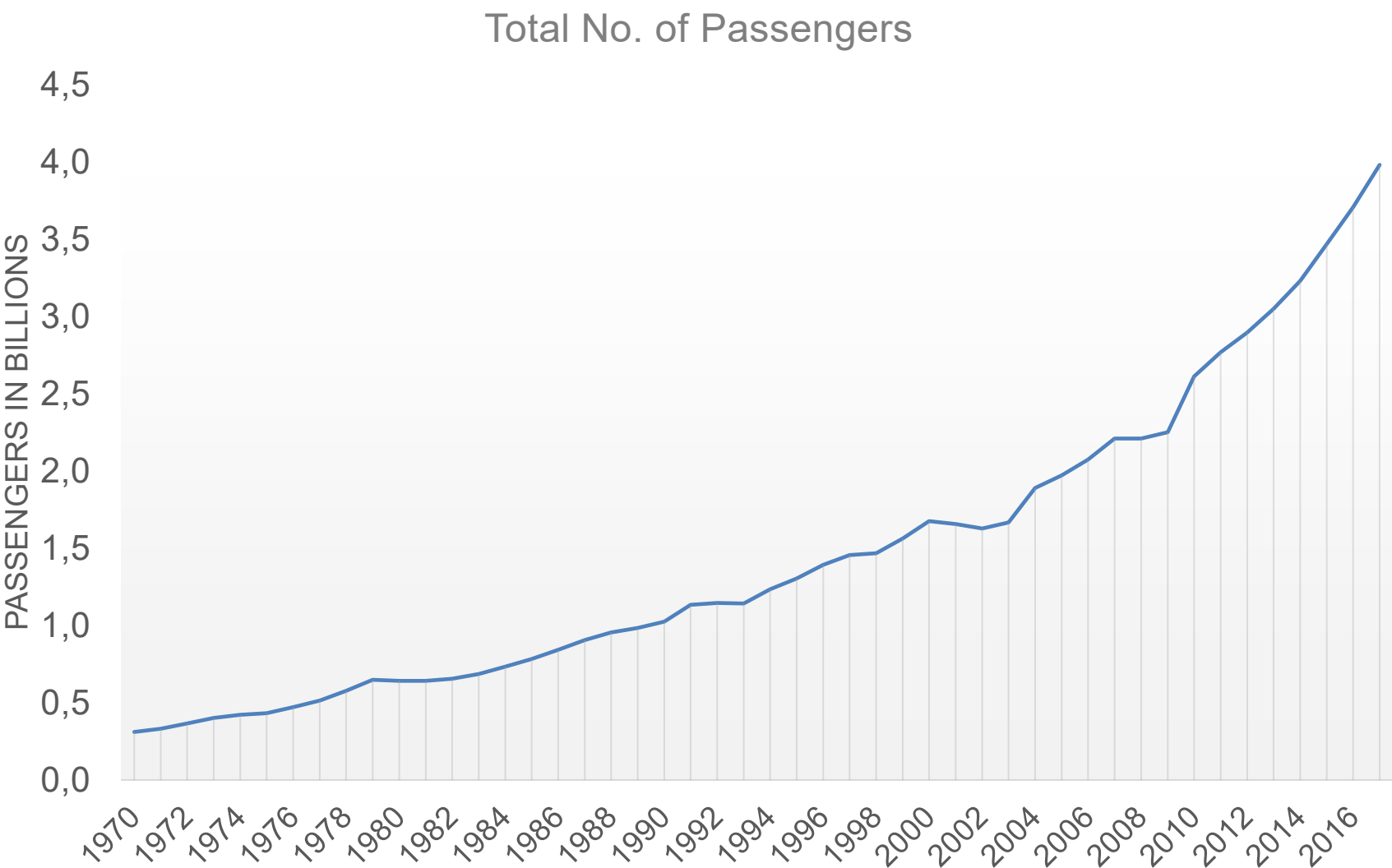
How do we travel

The energy requirements for mobility will continue to increase



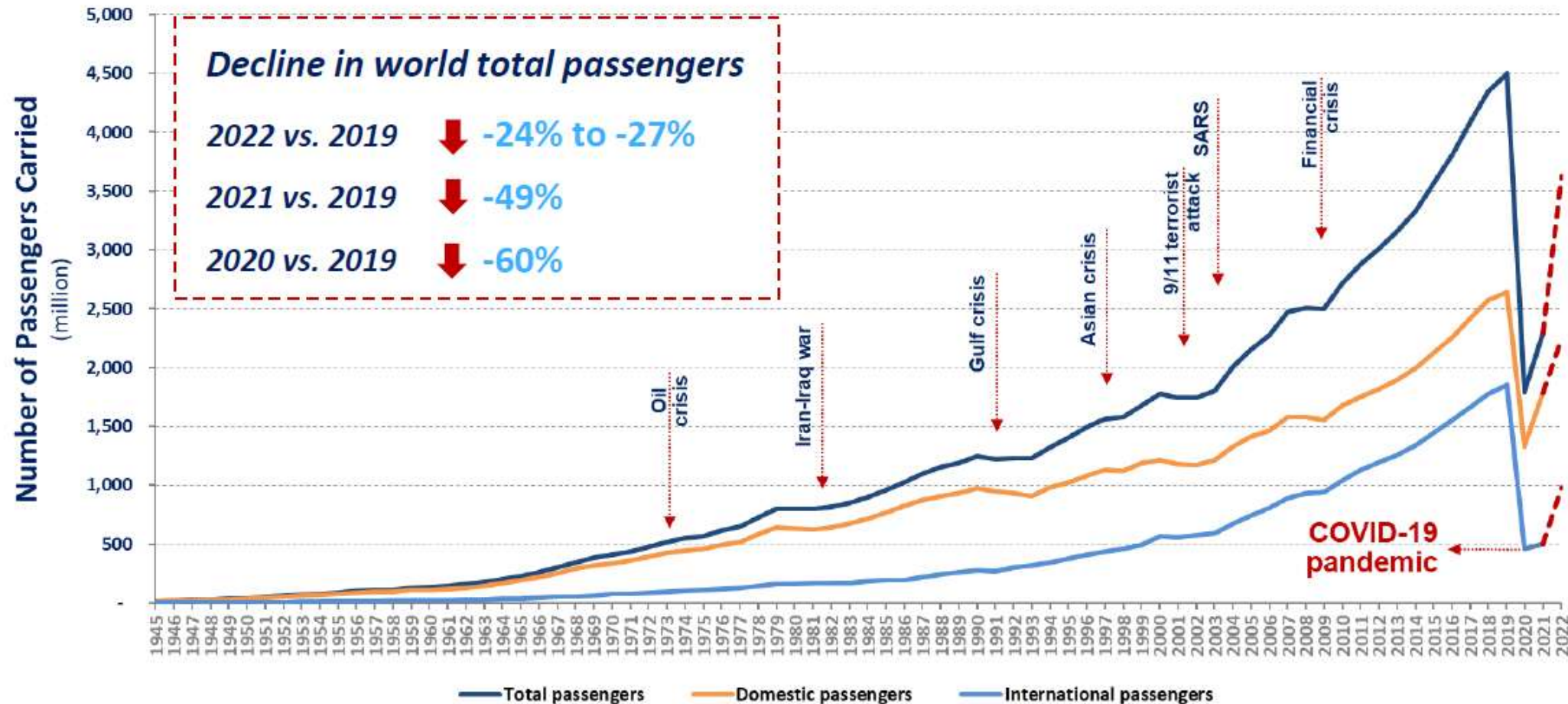
Inspiration from www.gapminder.org/ Dollar street

Total number of passengers travelling



The effect of Corona crisis

World passenger traffic evolution 1945 – 2022



Air traffic in a day (2005)



Please note that the air traffic today is more than double!!

The European Airspace

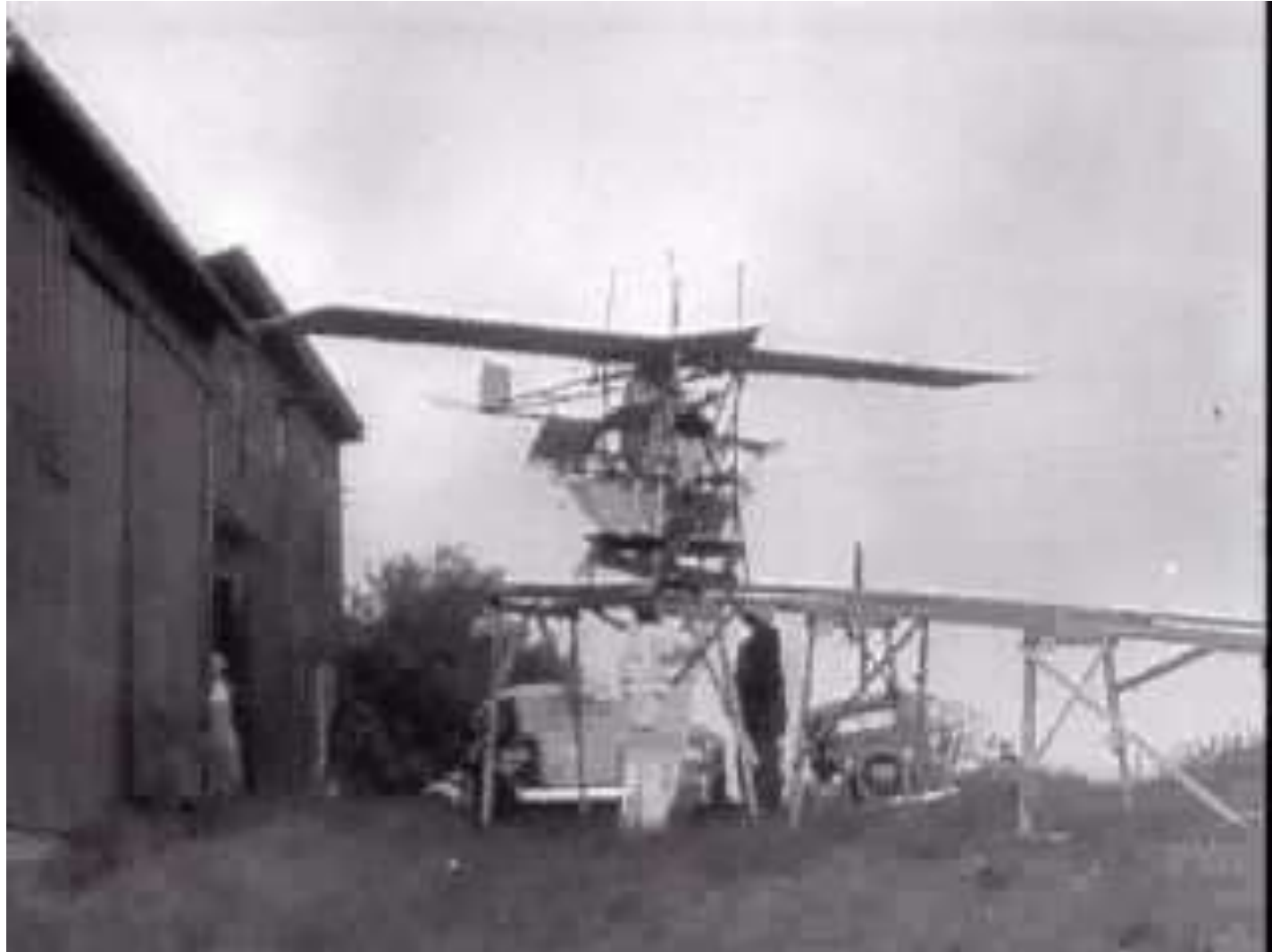
The image shows the NATS logo, which consists of the word "NATS" in a white, stylized, sans-serif font. The logo is centered within a large black rectangular area.

The beginning

Early flying attempts

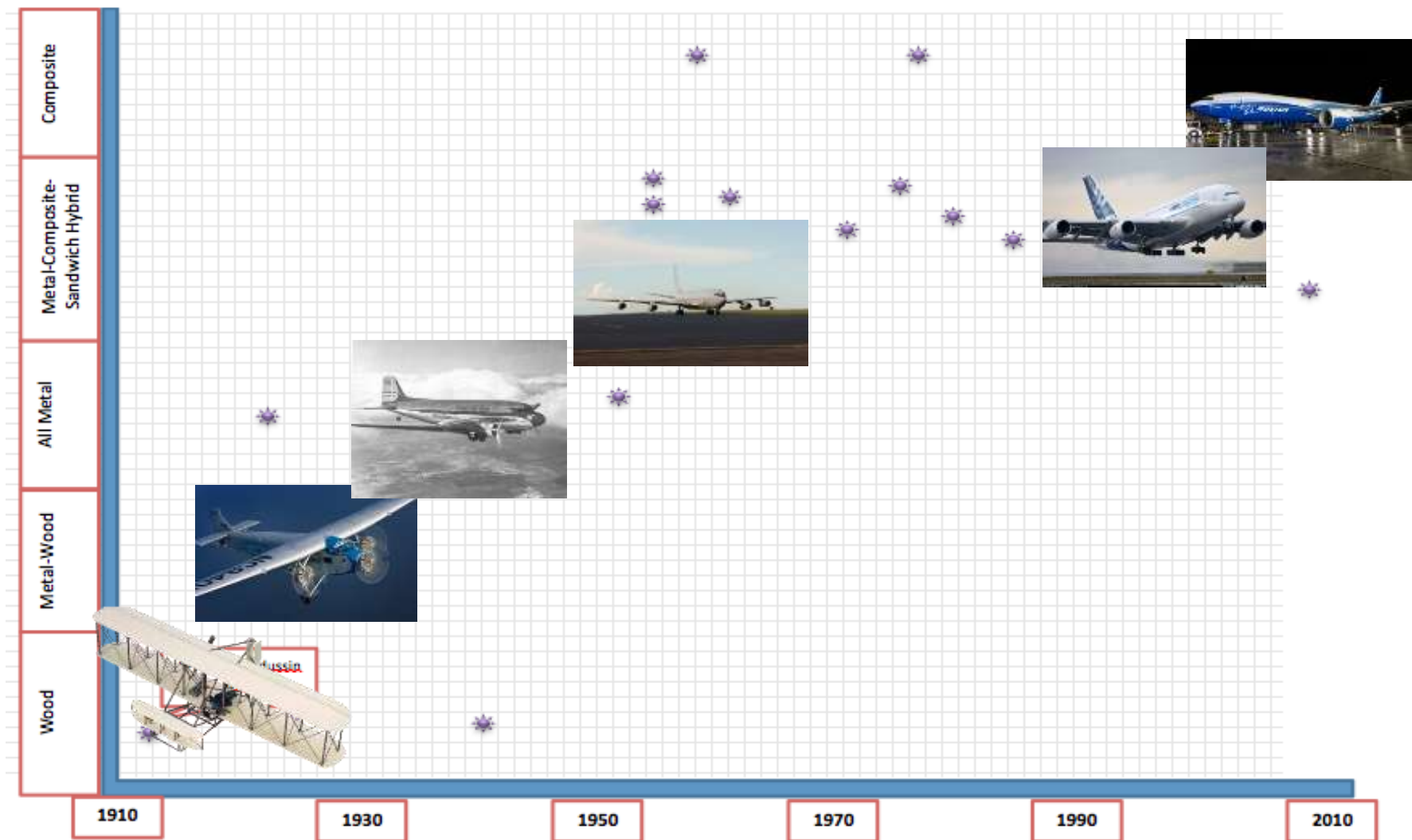


Early flying attempts



Evolution in Aircraft

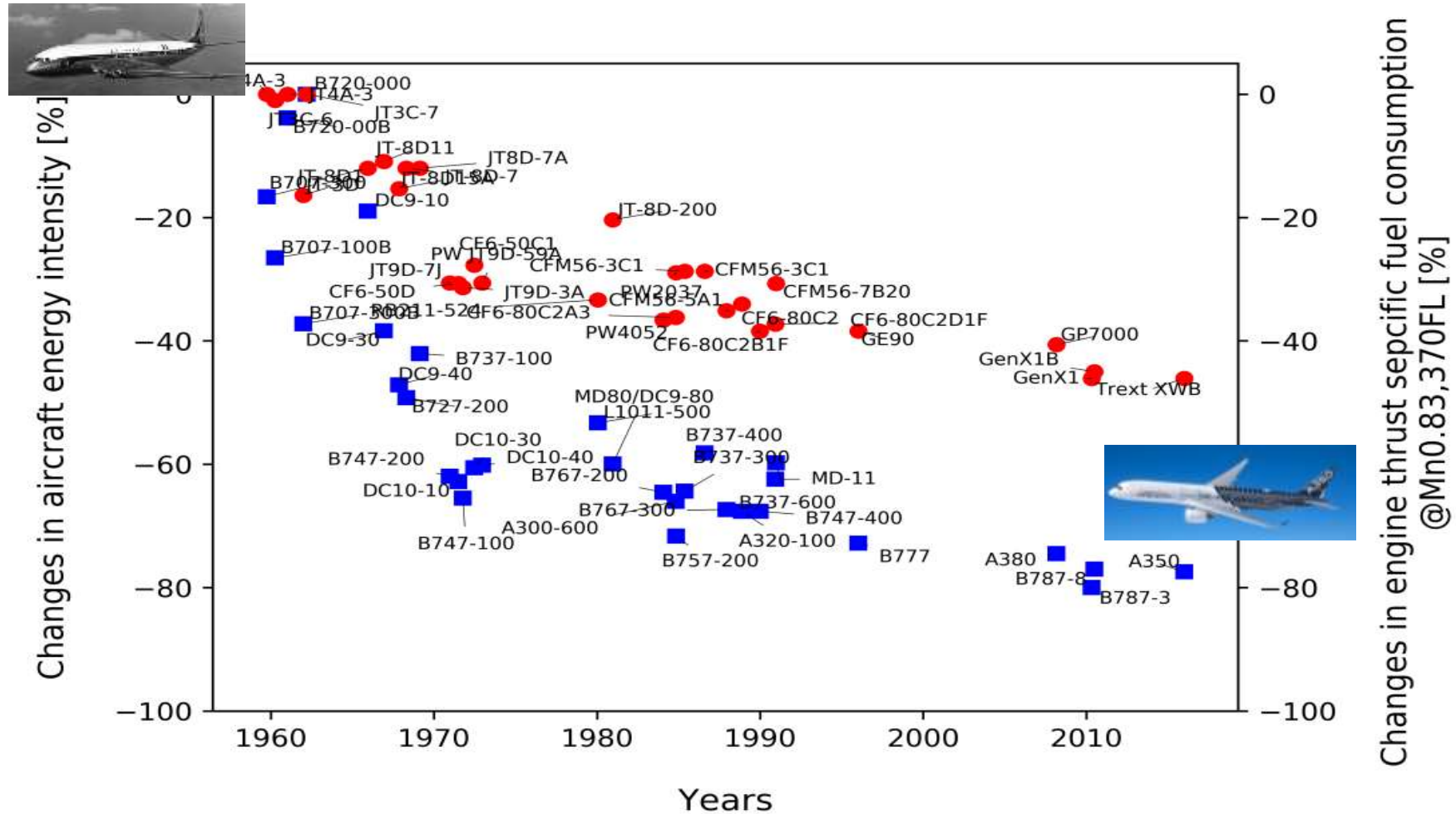
AE4238



Slide from Ronald van Gent

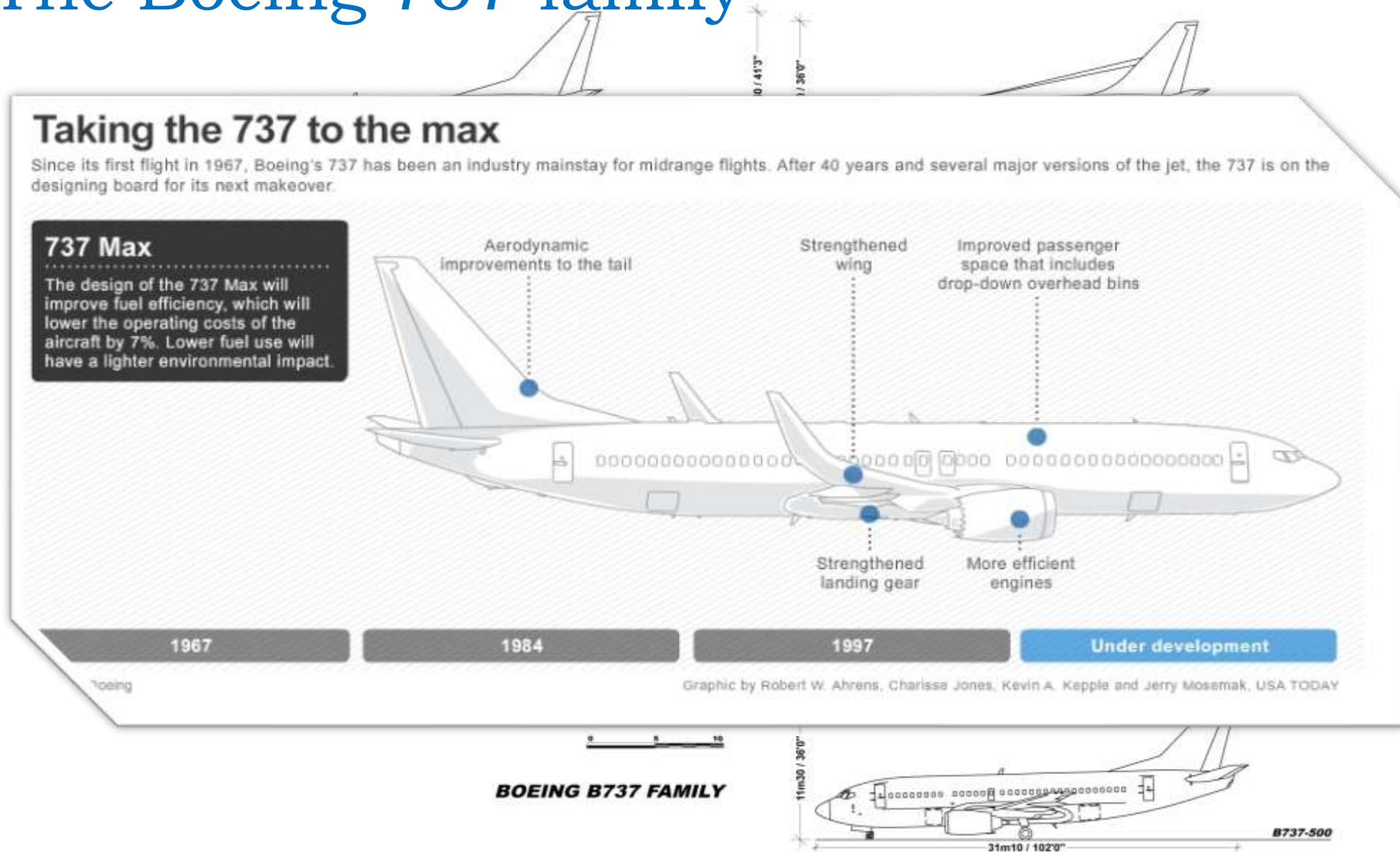
AE4238

Aircraft fuel burn



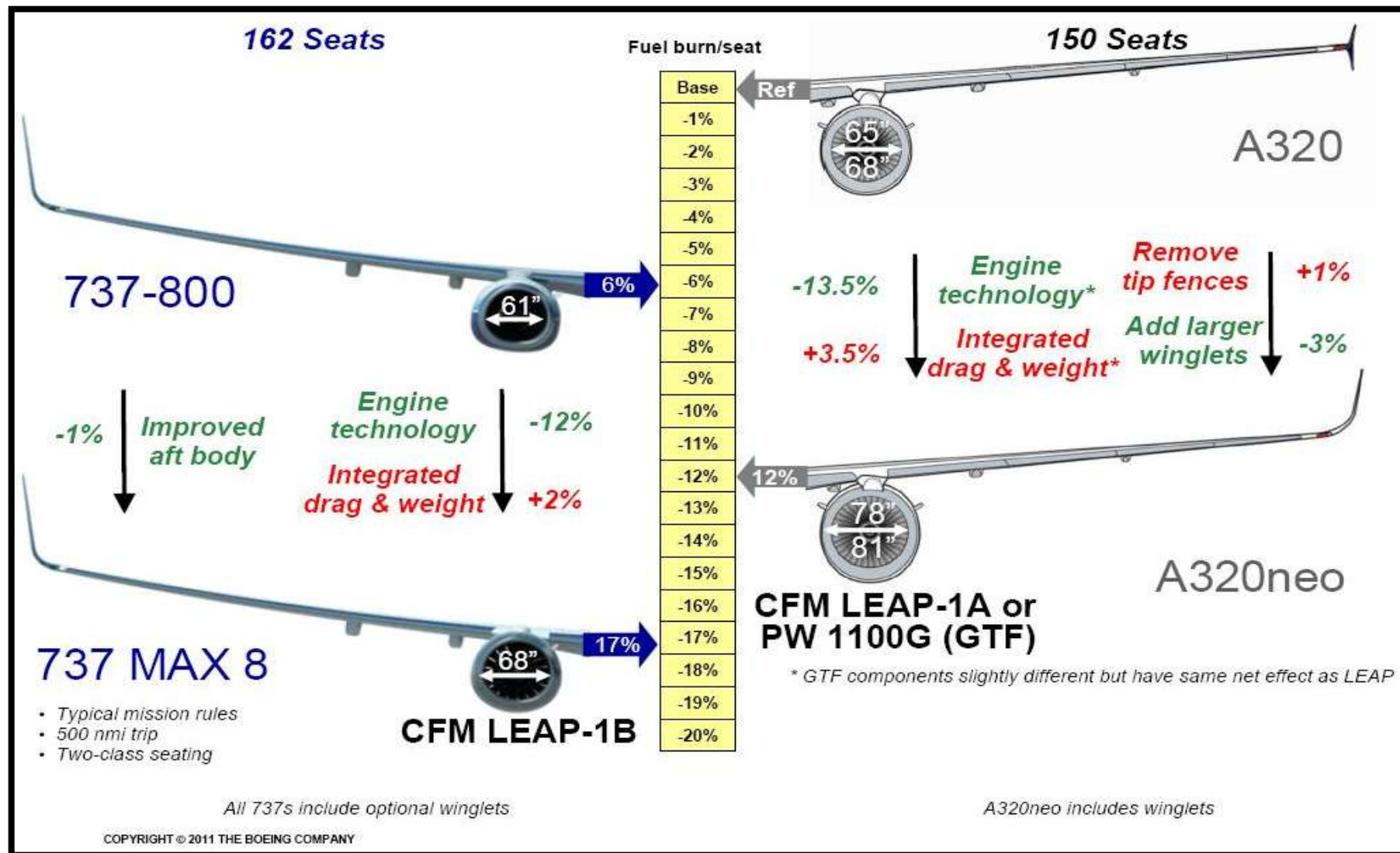
F. Yin & A. Gangoli Rao "A Review of Inter-stage Turbine Burner Turbofan Engine Concept for Future Civil Aviation"

The Boeing 737 family

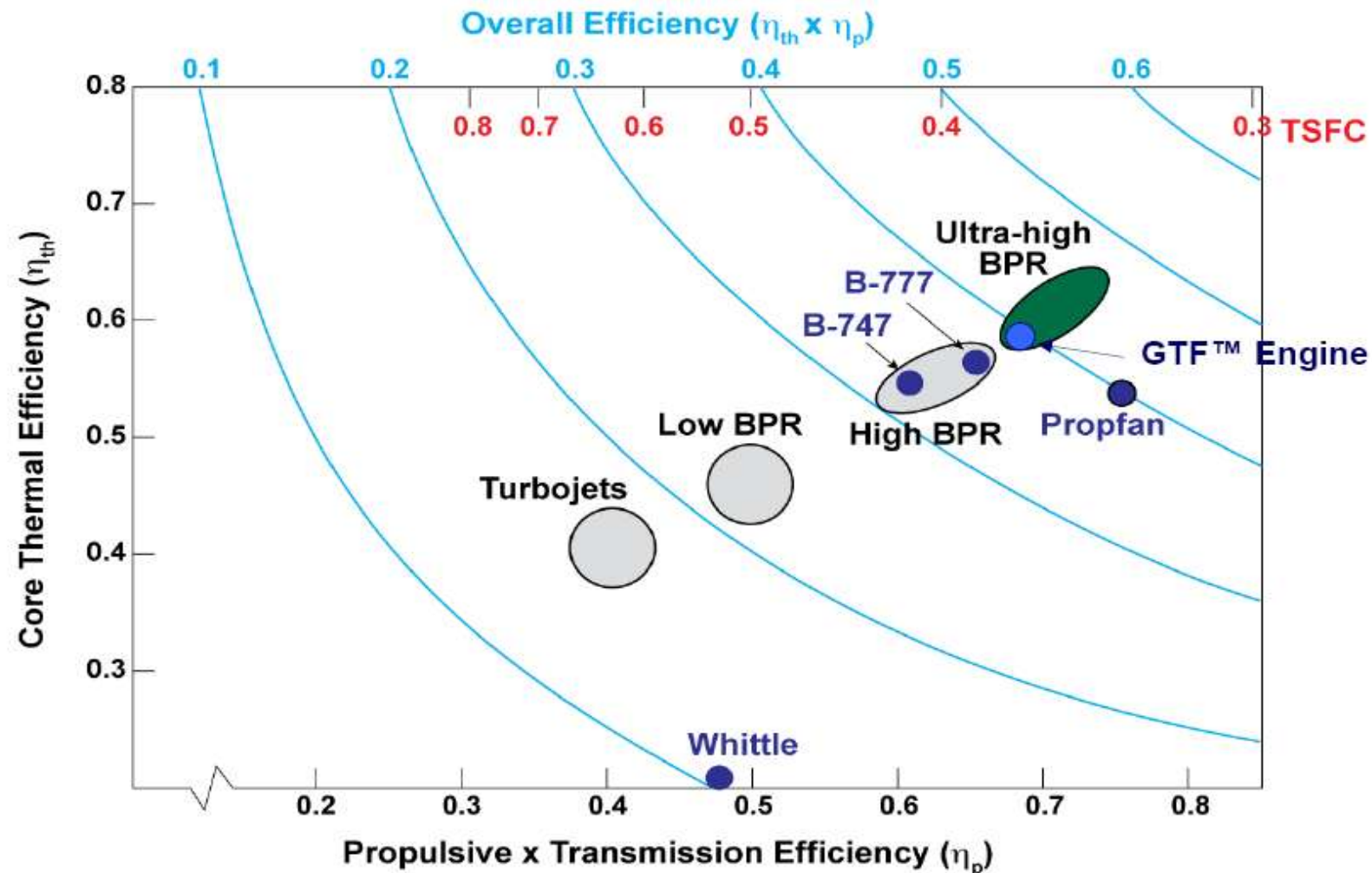


AE4238

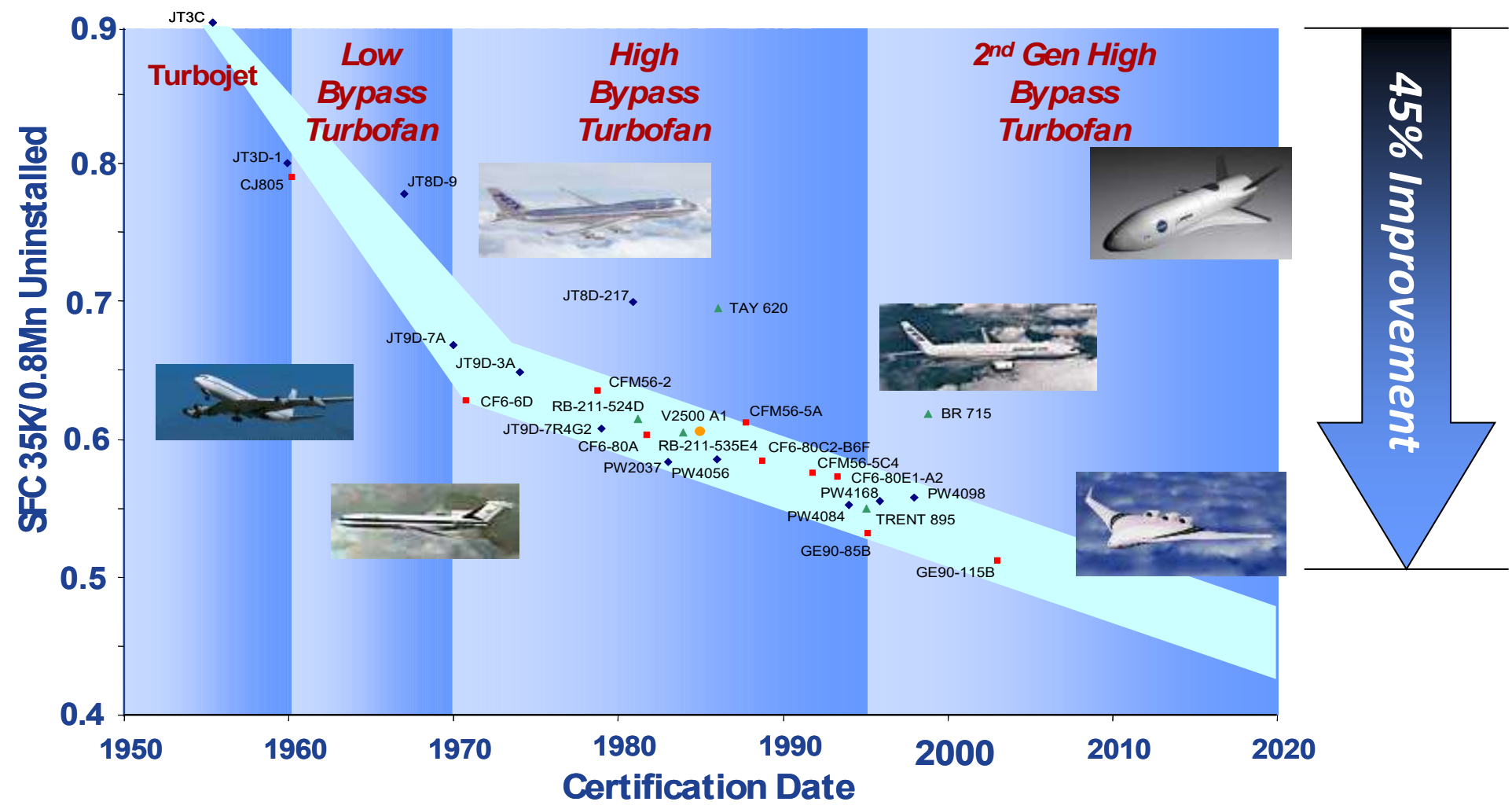
A commercial slide from Boeing



Thermal and Propulsive Efficiency Key to Reducing Specific Fuel Consumption (TSFC)

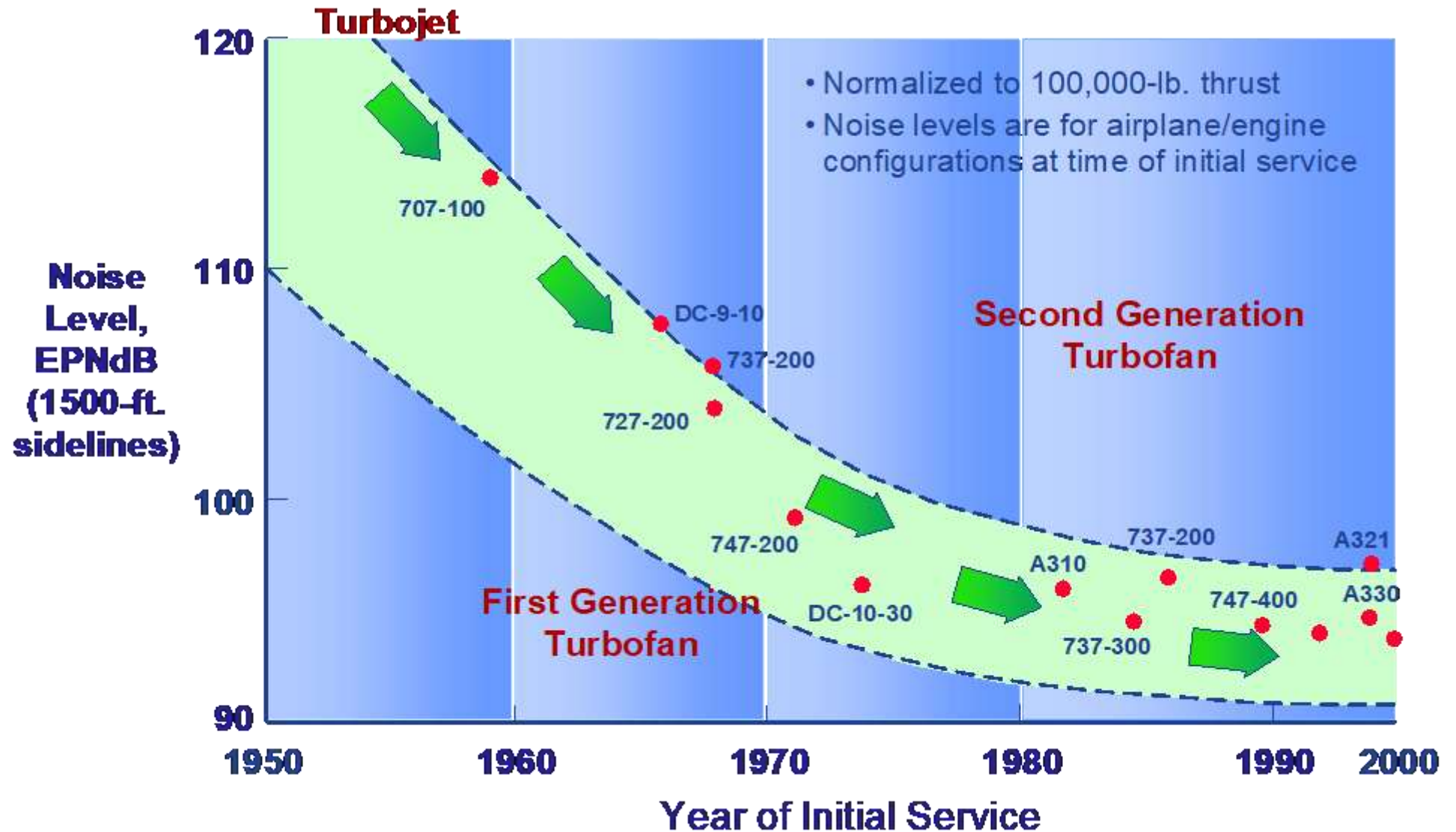


Reduction in engine SFC



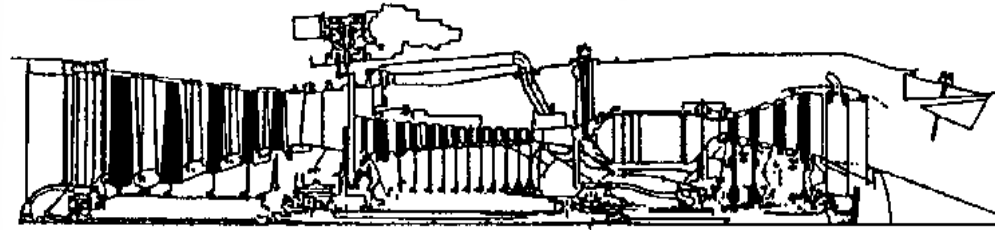
Reduction in Aircraft Noise

AE4238

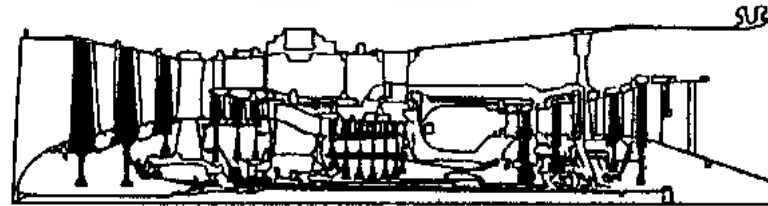


Thrust to Weight Ratio

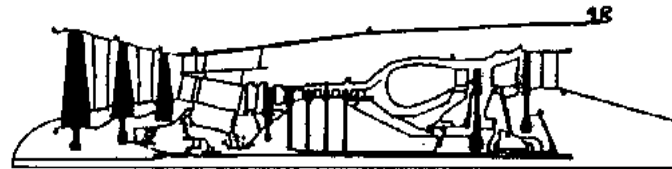
AE4238



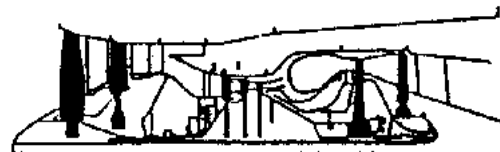
Spey 202



RB 199



EJ 200



Future

Engines scaled to equal thrust

Courtesy: Rolls Royce

Why Aero Engine Technology

“The best airplanes can only be designed around the best engines.”

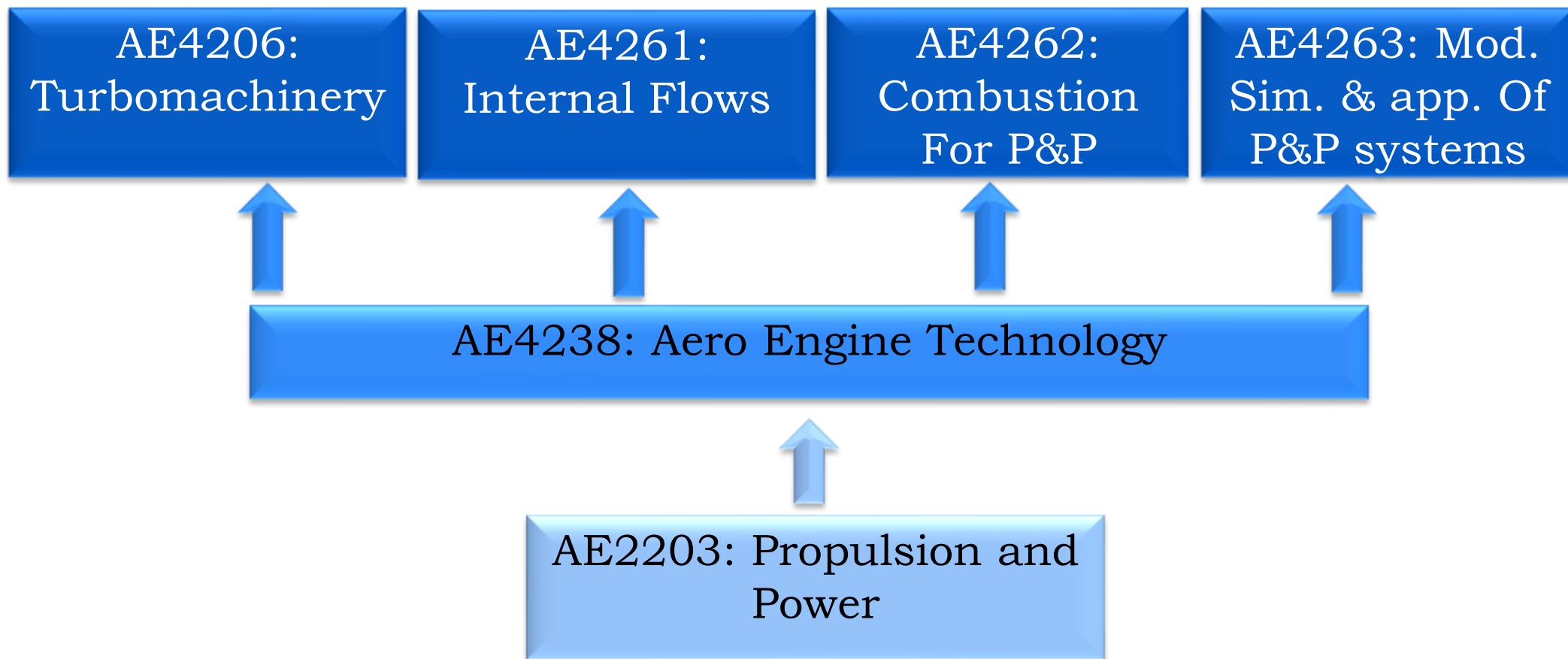
Frederick Rentschler

AE4238

AE4238

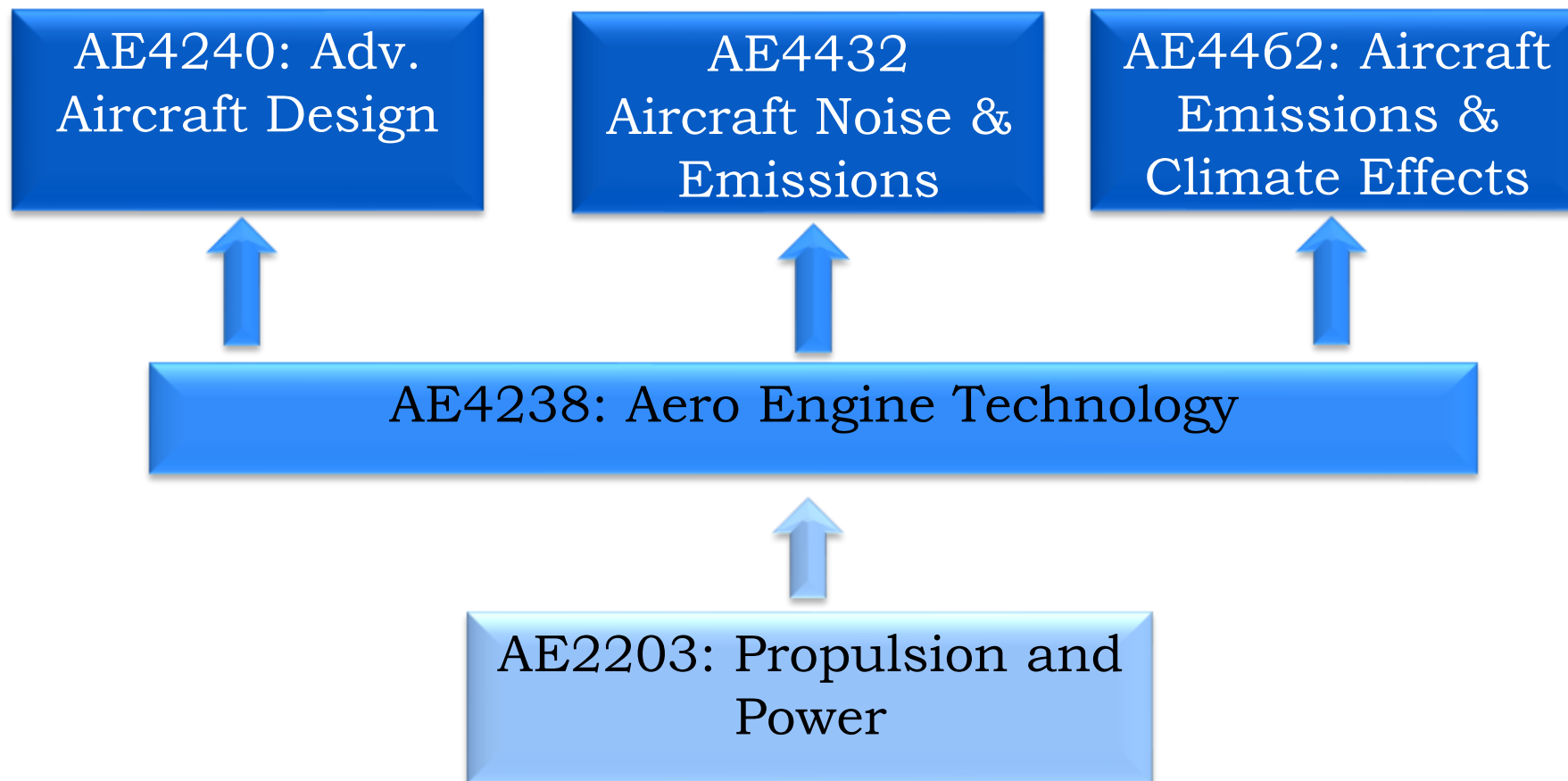
AE4238

Related Propulsion courses

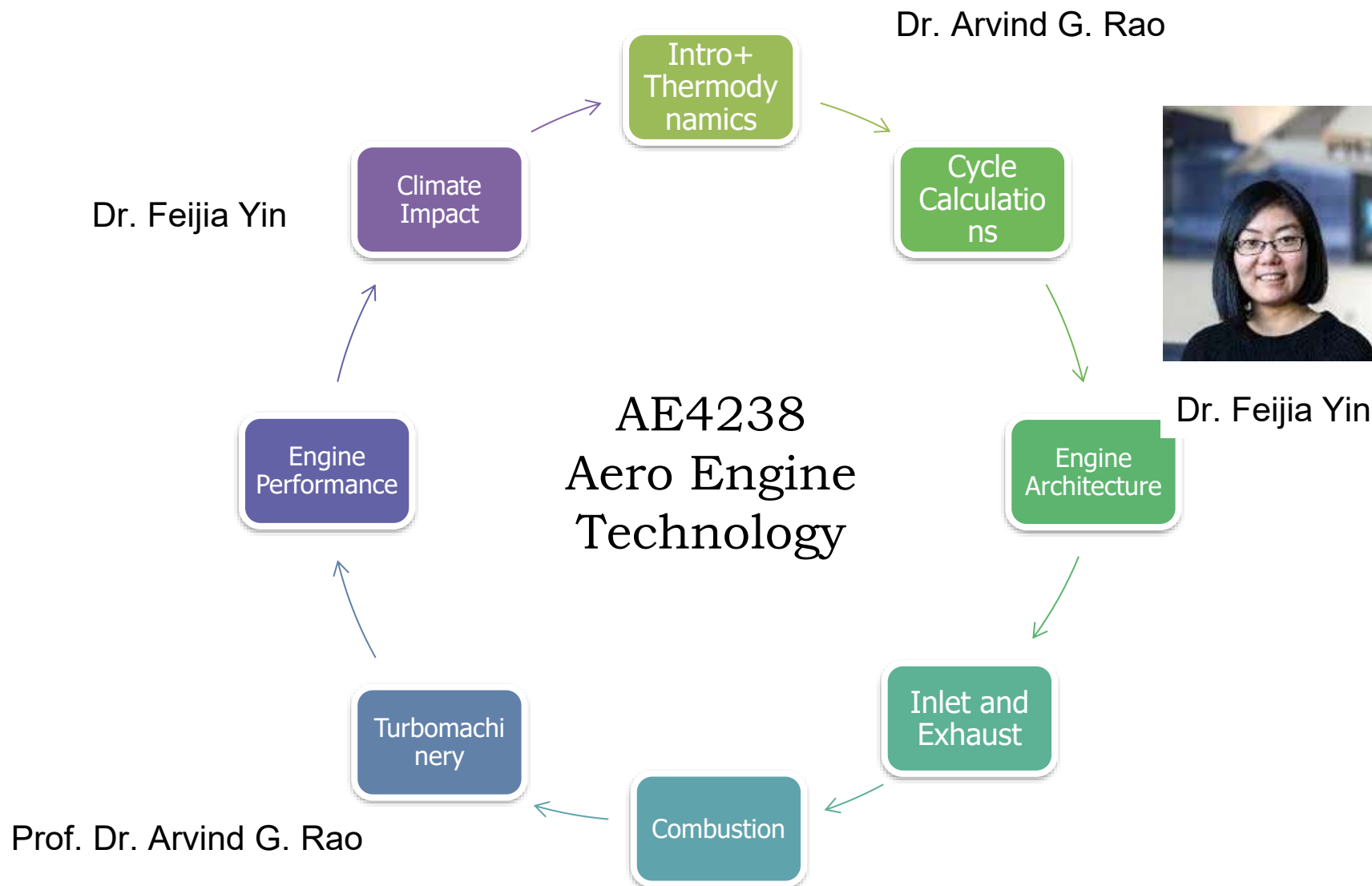


AE4238

Related Flight Performance courses



AE4238 Overview

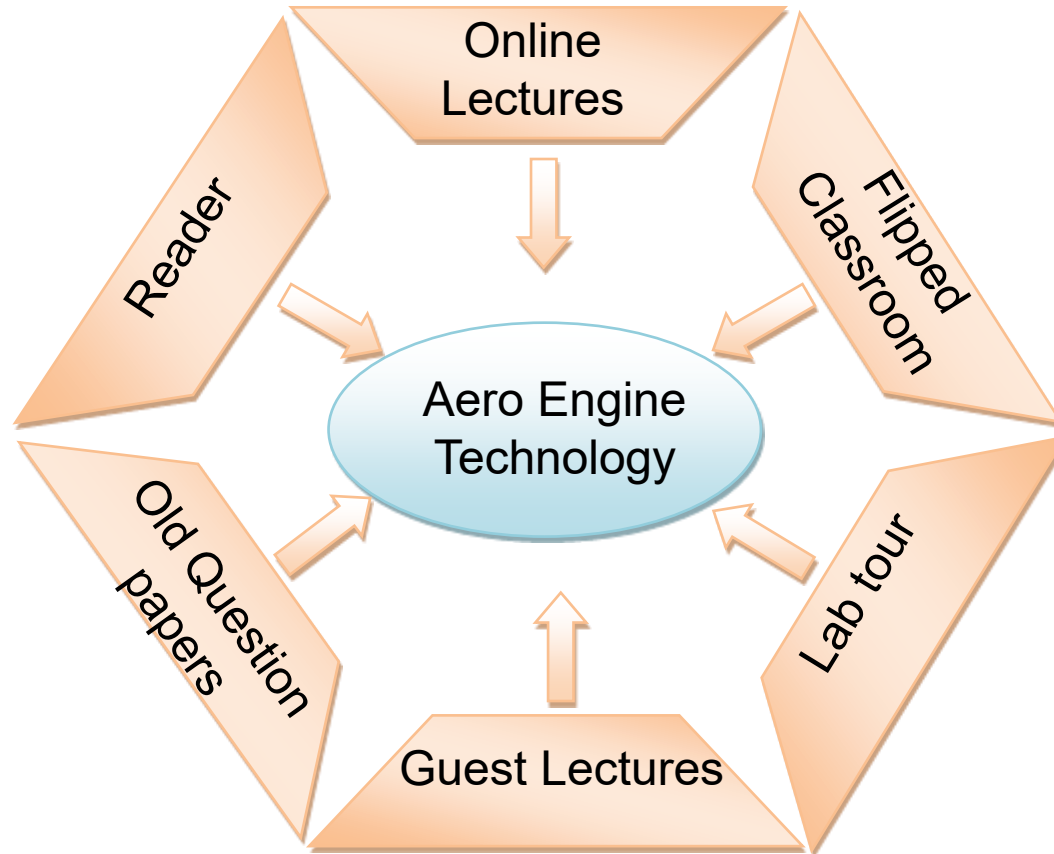


AE4238

Timetable 2025-26

AE4238	Week	Date	Day	ClassRoom	Time	Topic	Method	Assignment	Lecturer
Lecture-1	2.1	11-nov-25	Tuesday	AE-LR-CZ-G	8:45-10:30	Introduction & Thermodynamics	Inclass		Prof. dr. A. Gangoli Rao
Lecture-2	2.1	14-Nov-25	Friday	AE-LR-CZ-A	10:45-12:30	Cycle Efficiency & Cycle Calculations	Flipped		Dr. Feijia Yin
Lecture-3	2.2	18-Nov-25	Tuesday	AE-LR-CZ-G	8:45-10:30	Aero Engine Efficiency & Design Trends	Flipped	Assignment	Prof. dr. A. Gangoli Rao
Lecture-4	2.2	21-Nov-25	Friday	AE-LR-CZ-A	10:45-12:30	Engine Architecture-I	Flipped		Prof. dr. A. Gangoli Rao
Lecture-5	2.3	25-Nov-25	Tuesday	AE-LR-CZ-G	8:45-10:30	Engine Architecture-II	Flipped		Prof. dr. A. Gangoli Rao
Lecture-6	2.3	28-Nov-25	Friday	AE-LR-CZ-A	10:45-12:30	Engine Inlet System	Flipped		Prof. dr. A. Gangoli Rao
Lecture-7	2.4	2-Dec-25	Tuesday	AE-LR-CZ-G	8:45-10:30	Engine Exhaust System	Flipped		Prof. dr. A. Gangoli Rao
Lecture-8	2.4	5-Dec-25	Friday	AE-LR-CZ-A	10:45-12:30	Combustion-I	Flipped		Prof. dr. A. Gangoli Rao
Lecture-9	2.5	9-Dec-25	Tuesday	AE-LR-CZ-G	8:45-10:30	Combustion-II	Flipped	Assignment	Prof. dr. A. Gangoli Rao
Lecture-10	2.5	12-Dec-25	Friday	AE-LR-CZ-A	10:45-12:30	Turbomachinery-I	Flipped		Prof.dr. A. Gangoli Rao
Lecture-11	2.6	16-Dec-25	Tuesday	AE-LR-CZ-G	8:45-10:30	Turbomachinery-II	Flipped	Tutorials	Prof. dr. A. Gangoli Rao
Lecture-12	2.6	19-Dec-25	Friday	AE-LR-CZ-A	10:45-12:30	Performance and Control	Flipped		Prof. dr. A. Gangoli Rao
Lecture-13	2.7	6-Jan-26	Tuesday	AE-LR-CZ-G	8:45-10:30	Environmental Impact of Aviation	In Class		Dr. Feijia Yin
Lecture-14	2.7	9-Jan-25	Friday	AE-LR-CZ-A	10:45-12:30	Exam Questions	In Class	Tutorial	Prof. dr. A. Gangoli Rao

AE4238: Resources



Teaching Assistant

- Turhan Eker

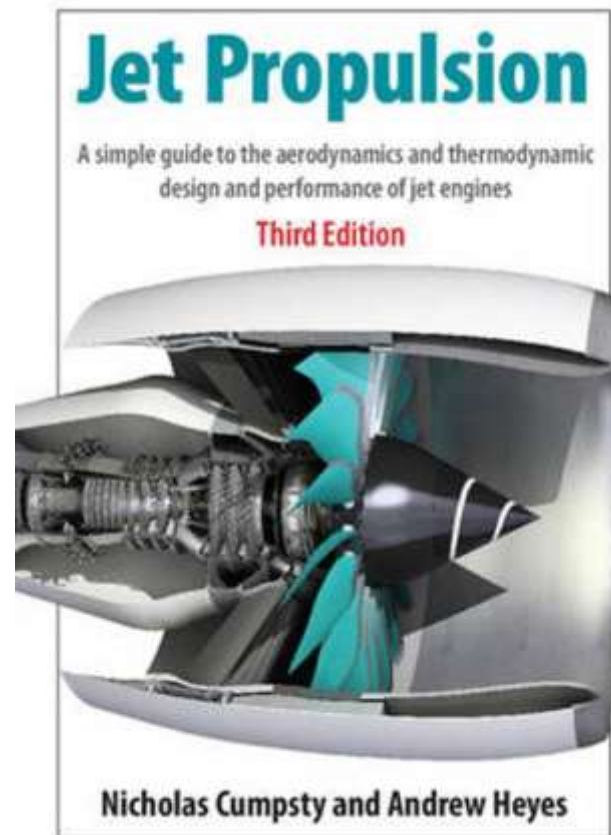
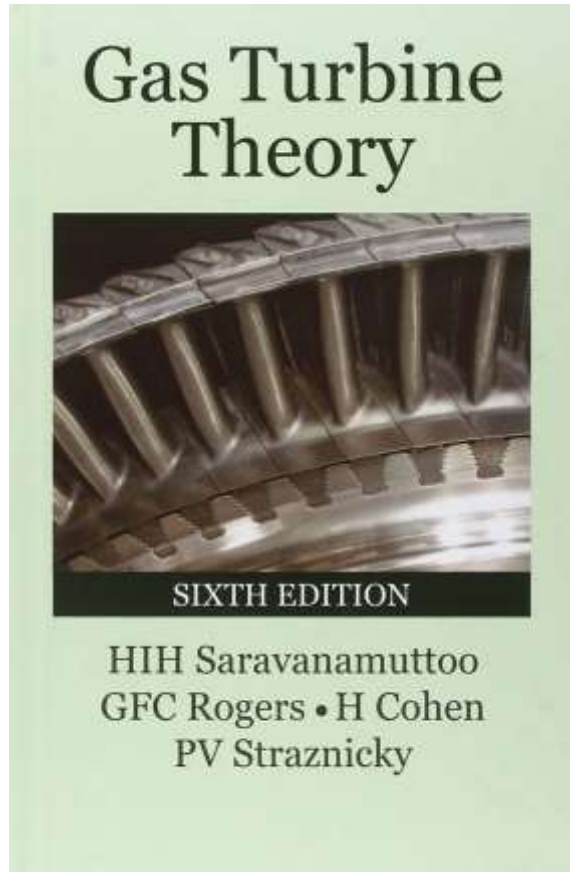


- Debducta Chatterjee



Please note that the reader does not contain everything that is taught in the class!!

Books (not mandatory)



AE4238

Lab Tour



Register for the tour in brightspace

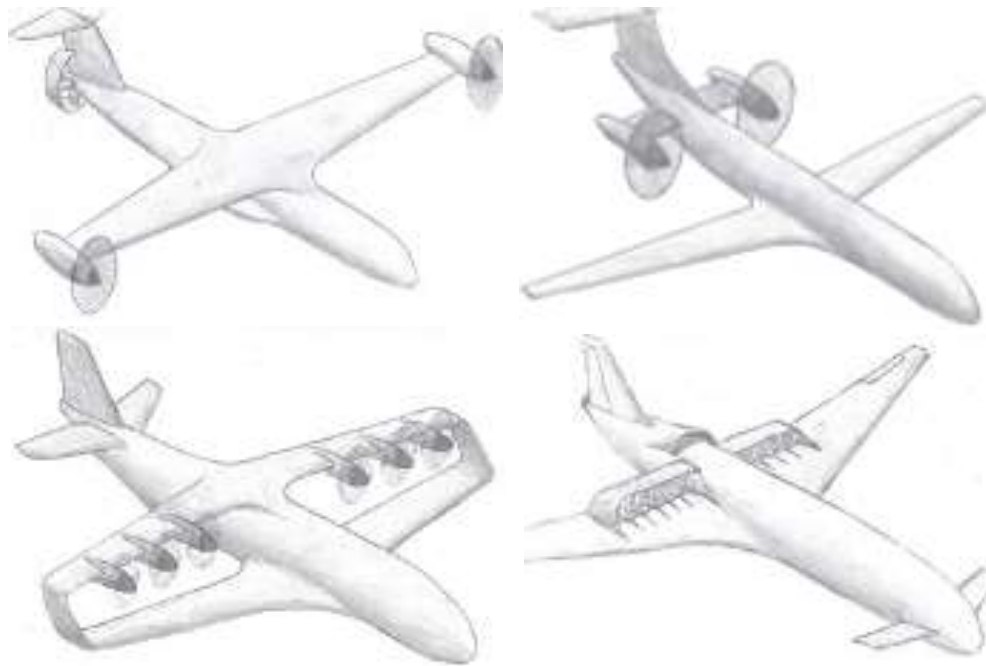
Important Points

- This is a 4 ECTS course.
- The primary mode of teaching would be flipped classroom.
- ***Flipped classroom is not an online course***
- Most of the Lectures have been recorded and available on BrightSpace.
- Students should go through the online-lecture before coming to the classroom.
- During the lecture we will discuss your doubts and questions.
- Lectures will follow the schedule mentioned on the timetable
- Please ask questions to the instructors. We encourage interaction!
- The TA's will have an open hour to answer your questions and help you in your studies.
- This is an MSc course, thus the course is shaped by you, we are just facilitators 😊

Assessment

- The grading shall be based on a final exam + Assignments.
- Overall, the division of grade between the assignments and the final exam is 30-70.
- However, you have to get a minimum of 5.5 grade points in the final exam to pass the course. So, if you do well in the assignments but fail to get 5.5 grade points in the final exam, it will still be considered as FAIL.
- There will be around 2 assignments in total.
- You might be randomly called by the instructor for a discussion w.r.t. the assignments.
- Please note that plagiarism is taken seriously in TU, so avoid copying or cheating in any way

Thank You



[Dr. Arvind Gangoli Rao](#)
a.gangolirao@tudelft.nl

